Video: Pearson's Correlation, Clearly Explained!!!

Video Link: https://youtu.be/xZ_z8KWkhXE
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I. The Core Concept: What is a Relationship?

- **Topic:** This video (Part 2 of 2) explains **Pearson's Correlation**, which builds directly on Covariance [00:17].
- **Example Scenario:** We are comparing paired data:
    i. mRNA transcripts for **Gene X** (on the x-axis) from 5 cells.
    ii. mRNA transcripts for **Gene Y** (on the y-axis) from the *same* 5 cells [00:55].
    - (Analogy: Counting green apples vs. red apples in the same 5 grocery stores) [01:14].
- **Visualizing the Trend:** We plot these pairs as dots. We can see a trend: cells with low Gene X also have low Gene Y, and cells with high Gene X also have high Gene Y [01:34].
- **Using the Trend Line:** We can draw a straight line (a "trend line") to represent this relationship [01:50]. This line allows us to make predictions or "educated guesses."
    - If we find a new cell with Gene X = 20, we can use the line to predict that its Gene Y value will be around 27 [01:58].
- **Correlation vs. Causation (Crucial Point):**
    - Just because we observe a trend **does not mean** one variable *causes* the other [03:59].
    - We are *not* saying high Gene X *causes* high Gene Y.
    - It's possible a third, unobserved factor is causing both genes to increase together [04:07]. Correlation simply describes the observed relationship.

II. Quantifying the *Strength* of a Relationship

- **What Correlation Measures:** Correlation quantifies the **strength** of a *linear* relationship [04:27]. It measures how well a straight line can describe the data.
- **Weak Relationship:** If the data points are far from the trend line, the relationship is "weak." This means a value for Gene X tells us very little about Gene Y (our predicted range is large) [03:01]. This results in a **small correlation value** (close to 0) [04:31].
- **Strong Relationship:** If the data points are *very close* to the trend line, the relationship is "strong." This means a value for Gene X can tell us a lot about Gene Y (our predicted range is very narrow) [02:32, 02:50]. This results in a **large correlation value** (close to 1 or -1) [04:45].
- **Independence from Scale:** Correlation **does not depend on the scale** of the data. The numbers on the axes do not matter [05:16]. It also doesn't matter what the slope is (steep or small); it only matters how well the points fit a *line* [05:41].

III. The Correlation Value (Range: -1 to +1)

- **Correlation = +1 (Perfect Positive Correlation):**
  - This is the maximum positive value [04:57].
  - It means a **perfect straight line** with a **positive slope** can go through the center of every single data point.
  - This represents the strongest possible *positive linear* relationship [18:03].
- **Correlation = -1 (Perfect Negative Correlation):**
  - This is the maximum negative value [09:50].
  - It means a **perfect straight line** with a **negative slope** can go through the center of every single data point.
  - This represents the strongest possible *negative linear* relationship [17:55].
- **Correlation = 0 (No Linear Correlation):**
  - This occurs when there is **no linear relationship** that can be represented by a straight line [11:29].
  - A value on the x-axis tells you *nothing* about what to expect on the y-axis [11:34, 18:28].
- **Values Between 0 and +/- 1:**
  - If a straight line *cannot* go through all the data points, the correlation value will move closer to 0 [11:15].
  - The "worse the fit" (the more scattered the points are), the closer the correlation gets to 0 [11:21].

IV. Confidence in Correlation (The p-value)

- This is a separate but critical concept: how much **confidence** do we have that the relationship we see is *real* and not just a random fluke?
- **The Small Sample Size Problem:**
  - If you have **only 2 data points** (n=2), you can *always* draw a perfect straight line between them [06:06].
  - This means with n=2, your correlation will *always* be +1 or -1, *even if the points are completely random* [06:44].
  - The relationship *appears* strong (correlation = 1), but we should have **no confidence** in it [06:16].
- **How Sample Size Increases Confidence:**
  - If you have 3 random points, the chance that they all fall on a perfect line is very small [07:31].
  - The more data points (n) you have that line up, the smaller the probability that it's just due to random chance [07:58, 08:07].
- **What the p-value Represents:**
  - A **p-value** for a correlation tells us the **probability that randomly drawn data would result in a relationship just as strong (or stronger)** as the one we observed [08:31].
  - **Small p-value (e.g., $p = 0.008$):** There is a very low probability that this is a random fluke. We have **high confidence** in the trend [12:44].
  - **Large p-value (e.g., $p = 0.8$):** There is a high probability (80%) that this trend is just random noise. We have **very little confidence** in it [12:26].
- **Key Distinction (Correlation vs. Confidence):**
  - You can have a **weak relationship** (correlation = 0.3) and **high confidence** (p-value = 0.008) at the same time [12:44].
  - This combination means you have a *lot of data* and you are *very confident* that the relationship is *truly weak* [13:06, 13:22].

- A small p-value does *not* make a weak relationship "good" for predictions. It just means the weak relationship is "real."

V. How to Calculate Correlation

- **Prerequisites:** You must first know how to calculate **Variance** and **Covariance** [13:39].
- **The Formula:**
  - The formula for Pearson's correlation (often denoted as r) is:

$$r = \frac{\text{Cov}(X,Y)}{\sqrt{\text{Var}(X)} \cdot \sqrt{\text{Var}(Y)}}$$

  - Or, written with standard deviations (s):

$$r = \frac{\text{Cov}(X,Y)}{s_X \cdot s_Y}$$

- **Why This Formula Works (The "Squeezing" Analogy):**
  - The **Covariance** (the numerator) is sensitive to the scale of the data. A covariance of 11.6 could be large or small, we don't know [14:13].
  - The **Denominator** (the product of standard deviations) accounts for the scale and variance of *both* variables.
  - By dividing, the denominator **"squeezes" or normalizes** the covariance, canceling out the effect of scale [14:38]. This forces the final value to be within the standard range of **-1 to 1**, making it interpretable [14:47].
- **Example Calculation:**
  - From the video (using data from the previous Covariance StatQuest):
  - $\text{Cov}(X,Y) = 11.6$ [15:29]
  - $\text{Var}(X) = 101.8$ [15:46]
  - $\text{Var}(Y) = 160.3$ [15:46]

$$r = \frac{11.6}{\sqrt{101.8} \cdot \sqrt{160.3}} \approx 0.9$$

  - **Correlation (r) = 0.9** [15:55]
  - The p-value for this result was 0.03 (a 3% chance the relationship is random) [16:17].

VI. The Problem with Correlation: R-Squared

- Even correlation values are "not super easy to interpret" [16:38].
- For example, is a correlation of $r = 0.9$ "twice as good" for making predictions as a correlation of $r = 0.64$? It's not obvious [16:47].
- The solution to this interpretation problem is **R-squared (R^2)**, which is the topic of another StatQuest [17:03].
- R^2 is also useful because it can quantify relationships that are more complex than a simple straight line [17:20].

http://googleusercontent.com/youtube_content/4